

1.4 Developments in modern and smart materials, composite materials and technical textiles	To apply technical knowledge and understanding of the characteristics, applications, advantages and disadvantages of the following.
	1.4.1 Modern and smart materials: - a shape-memory alloys (SMAs) - b nanomaterials - c reactive glass - d piezoelectric materials - e temperature-responsive polymers - f conductive inks.
	1.4.2 Composites: - a concrete - b plywood - c fibre/carbon/glass - d reinforced polymers - e robotic materials.

CORE

Key idea	What students need to learn:
	1.4.3 Technical textiles: - a agro-textiles - b construction textiles - c geo-textiles - d domestic textiles - e environmentally friendly textiles - f protective textiles - g sports textiles.